

Venkata Pavan Potturu

Software Developer

CO, USA | venkatapotturu@itjobtech.com | 720-341-7034 | [LinkedIn](#)

SUMMARY

- Designed and developed scalable **software solutions** over **4+ years**, employing **Java, Python, C#, JavaScript**, and **SQL** to create efficient **backend systems** that improved processing speed by an average of **30%**.
- Architected high-performing software solutions utilizing advanced **Data Structures and Algorithms**, achieving a **30%** increase in **system efficiency** while ensuring maintainability through **OOP principles** like **Encapsulation and Polymorphism**.
- Proficient in Full-Stack Development & Cloud Technologies**, utilizing **Spring Boot, Hibernate, React.js, Angular, AWS (EC2, S3, Lambda), and Azure (SQL, Functions)** to deploy and manage microservices-based, event-driven applications.
- Strong in **API Development, DevOps, and Agile Methodologies**, designing and securing **RESTful APIs, GraphQL, and gRPC** while implementing **CI/CD pipelines (Jenkins, GitHub Actions), Docker, Kubernetes**, and collaborating in **Agile/Scrum environments** for high-quality software delivery.

EXPERIENCE

Software Developer, Berkshire Hathway | USA

Jan 2024 – Present

- Engineered a well-architected inventory management system using **Java, C#, and React.js**; implemented **OOP principles** such as **Encapsulation, Inheritance, and Polymorphism** to enhance maintainability by **40%** compared to previous systems.
- Designed and optimized **RESTful APIs**, ensuring **99.9% uptime** and improving API response times by **50%** through efficient **Data Structures & Algorithms**.
- Orchestrated cloud-based deployment using **AWS (EC2, S3, Lambda, API Gateway)**, enhancing scalability and availability by **30%**.
- Integrated **core design principles** of **MVC and Factory patterns** into existing **frameworks**; streamlined overall software architecture leading to a reduction in bugs during deployment cycles by over **30%**.
- Established resilient **authentication mechanisms**, including **OAuth and JWT**, to secure **API endpoints**; enhanced **data protection** protocols resulting in a **40%** reduction in unauthorized access attempts across the platform.

Software Engineer, TCS | India

Jun 2021 – Jun 2022

- Developed critical modules for a high-traffic **CRM platform** using best practices in **software development**; findings from code reviews revealed three major areas causing inefficiencies that were subsequently addressed through targeted improvements.
- Constructed and deployed **SOAP APIs using Python** to facilitate legacy system integration, increasing data exchange efficiency by **20%**.
- Accelerated system performance by **leveraging Redis caching**, reducing **API response time by 30%** and enhancing throughput.
- Refined **microservices** using established Design Patterns such as **MVC and Singleton**; improved overall code structure leading to an increase in reusability rates by over **25%**, streamlining future project integrations.
- Coordinated** daily stand-ups and retrospectives in **Agile environments** while employing tools such as **JIRA**; identified key workflow bottlenecks that led to a more efficient sprint process without sacrificing quality standards.

Software Developer, Cyient | India

Apr 2020 – May 2021

- Built and enhanced core **e-commerce platform** features using **Java, Spring Boot, and Angular.js**, embedding **OOP principles (Encapsulation, Polymorphism)** to improve modularity.
- Developed and deployed **secure RESTful APIs**, integrating third-party payment systems to enable seamless transactions.
- Executed and scaled cloud infrastructure using **AWS (EC2, S3, Elastic Beanstalk)** while **automating deployment pipelines** with **Jenkins and GitHub Actions**.
- Applied and structured Design Patterns (**Factory, MVC**) to **optimize application architecture** and enhance maintainability.
- Tested and debugged applications through **unit and integration tests (JUnit, Selenium)**, ensuring high-quality, well-structured software releases.

SKILLS

Programming: Java, Python, C#, JavaScript, SQL, C++

Software Development: Data Structures & Algorithms, OOP (Encapsulation, Inheritance, Polymorphism), Design Patterns (MVC, Singleton, Factory)

Frameworks & Libraries: Spring Boot, Hibernate, Django, Flask, Express.js, React.js, Angular.js

Databases & Cloud: MySQL, PostgreSQL, MongoDB, Redis, AWS (EC2, S3, Lambda), Azure (SQL, Functions)

DevOps & APIs: Docker, Kubernetes, Terraform, CI/CD (Jenkins, GitHub Actions), RESTful APIs, GraphQL, gRPC, OAuth, JWT

Testing & Security: JUnit, pytest, Selenium, OWASP Security, SSL/TLS, OAuth, SAML

System Design & Agile: Microservices, Event-Driven Architecture, Kafka, RabbitMQ, Scrum, Git, SDLC

EDUCATION

Masters of Science - Computer Science, **University of Colorado Boulder, USA**

May 2024

Bachelors of Technology - Computer Science, **GITAM University, India**

May 2021